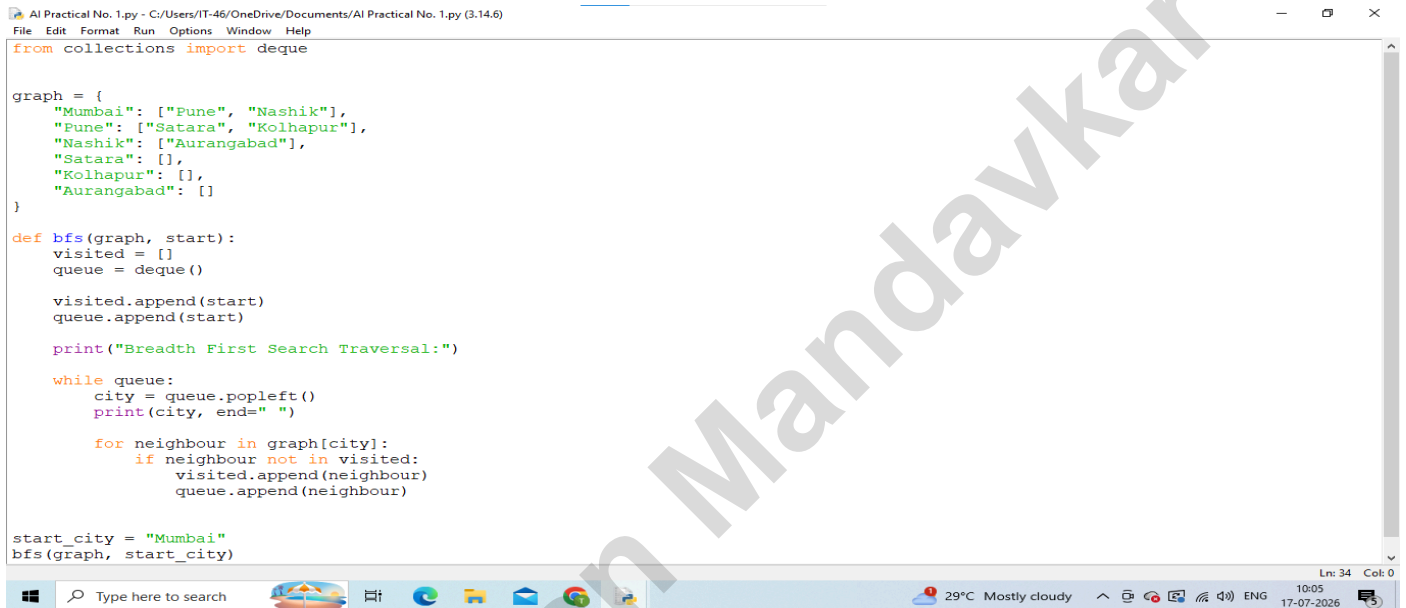


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Aim:- Breadth First Search & Iterative Depth First Search in Python.

A] Implement the Breadth First Search algorithm to solve a given problem

INPUT:-



```
from collections import deque

graph = {
    "Mumbai": ["Pune", "Nashik"],
    "Pune": ["Satara", "Kolhapur"],
    "Nashik": ["Aurangabad"],
    "Satara": [],
    "Kolhapur": [],
    "Aurangabad": []
}

def bfs(graph, start):
    visited = []
    queue = deque()

    visited.append(start)
    queue.append(start)

    print("Breadth First Search Traversal:")

    while queue:
        city = queue.popleft()
        print(city, end=" ")

        for neighbour in graph[city]:
            if neighbour not in visited:
                visited.append(neighbour)
                queue.append(neighbour)

start_city = "Mumbai"
bfs(graph, start_city)
```

OUTPUT:-



```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> ===== RESTART: C:/Users/IT-46/OneDrive/Documents/AI Practical No. 1.py =====
>>> Breadth First Search Traversal:
>>> Mumbai Pune Nashik Satara Kolhapur Aurangabad
```

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B) Implement the Iterative Depth First Search algorithm to solve the same problem.

INPUT:-

```
AI Practical No. 1.py - C:/Users/CHETAN/Documents/AI Practical No. 1.py (3.12.4)
File Edit Format Run Options Window Help

graph = {
    "Mumbai": ["Pune", "Nashik"],
    "Pune": ["Satara", "Kolhapur"],
    "Nashik": ["Aurangabad"],
    "Satara": [],
    "Kolhapur": [],
    "Aurangabad": []
}

def dfs(graph, start):
    visited = []
    stack = []

    stack.append(start)

    print("Iterative Depth First Search Traversal:")

    while stack:
        city = stack.pop()

        if city not in visited:
            print(city, end=" ")
            visited.append(city)
```

```
AI Practical No. 1.py - C:/Users/CHETAN/Documents/AI Practical No. 1.py (3.12.4)
File Edit Format Run Options Window Help

def dfs(graph, start):
    visited = []
    stack = []

    stack.append(start)

    print("Iterative Depth First Search Traversal:")

    while stack:
        city = stack.pop()

        if city not in visited:
            print(city, end=" ")
            visited.append(city)

            for neighbour in reversed(graph[city]):
                if neighbour not in visited:
                    stack.append(neighbour)

start_city = "Mumbai"
dfs(graph, start_city)
```

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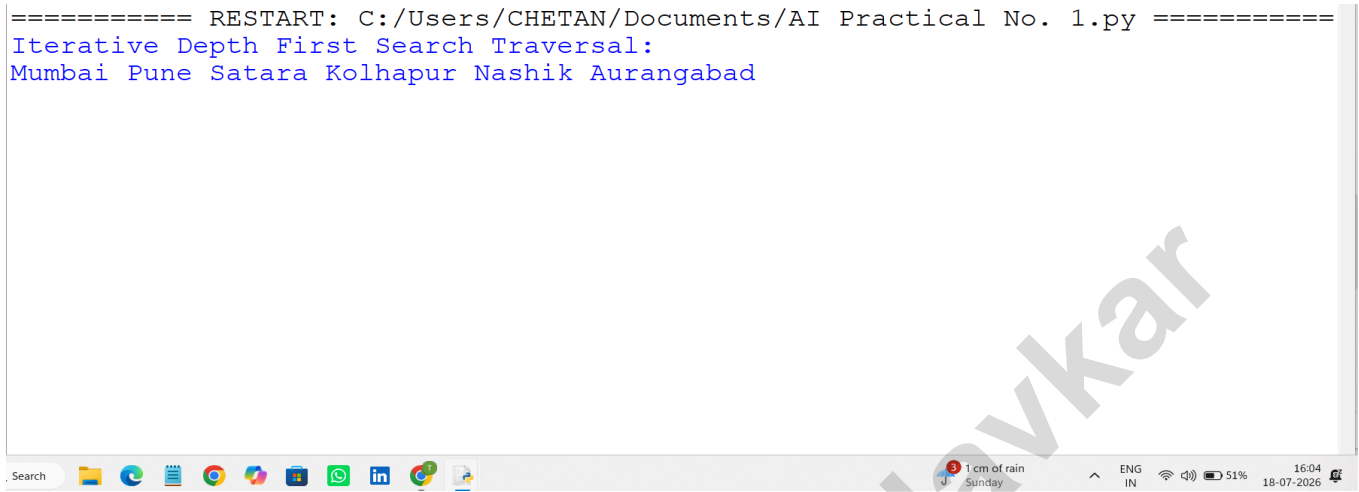
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OUTPUT:-

```
===== RESTART: C:/Users/CHETAN/Documents/AI Practical No. 1.py =====  
Iterative Depth First Search Traversal:  
Mumbai Pune Satara Kolhapur Nashik Aurangabad
```



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